

Haoxuan LI

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Education

Georgia Institute of Technology

August 2026-Present

- **Program:** Master of Science in Computer Science

The Hong Kong Polytechnic University

September 2022-May 2026

- **Program:** Bachelor of Science (Hons) in Computer Science | GPA: 3.83/4.3

- **Honour:**

2023-2026 Dean's Honour List of The Hong Kong Polytechnic University

October 2023-May 2026

Skills

- **Programming Languages:** Python, Java, SQL
- **Libraries & Frameworks:** PyTorch, PySyft, pandas, Java Swing, JUnit
- **Developer Tools:** Git
- **Languages:** English (Professional Working Proficiency), Chinese (Native)

Experience

Data Analyst Intern, Sina Weibo (China's Leading Social Media Platform)

May 2024-August 2024

- Built SQL queries and Python/pandas scripts to extract, clean, and standardize engagement and sentiment data, automating weekly reporting workflows.
- Enhanced proprietary analytics dashboards by configuring filters, validating outputs, and visualizing engagement, sentiment, topic trends, and campaign performance.
- Analyzed 10+ client accounts to identify emerging topics and performance shifts, translating findings into actionable reports and presentations for client-facing teams.

Research Assistant, The Hong Kong Polytechnic University

February 2025-August 2025

Building Climate Expert AI Agent with Retrieval-Augmented Generation (RAG)

- Developed a Python-based web crawler to collect and preprocess over 100 IPCC reports, transforming the extracted content into structured JSON data for a domain-specific climate knowledge base.
- Developed a DeepSeek-R1 question-answering agent and RAG pipeline using prompt engineering, vector embeddings, and semantic search to ground responses in authoritative reports.
- Evaluated generated responses using BLEU, ROUGE, and F1-score, and iteratively refined prompting and retrieval strategies, achieving a 10% improvement in aggregate evaluation scores.

Research Assistant, The Hong Kong Polytechnic University

February 2024-July 2024

Privacy-Preserving Federated Learning with Noisy Homomorphic Encryption

- Implemented privacy-preserving federated learning workflows using the PySyft framework and PyTorch, focusing on secure aggregation and encrypted model updates.
- Applied the BFV homomorphic encryption scheme to enable encrypted computation on distributed model parameters and ensure data confidentiality.
- Integrated noise-based privacy mechanisms into the federated learning pipeline and added random noise to model updates to reduce information leakage and enhance user privacy.
- Supported GAN-based adversarial experiments and evaluated the robustness of privacy-preserving methods using Attack Success Rate (ASR), classification accuracy, F1-score, and accuracy degradation under attack.

Projects

Full-Stack Monopoly Game Development

October 2024-December 2024

- Led a team to design and implement a local Monopoly game in Java and utilized the Model-View-Controller (MVC) architecture to ensure modularity, scalability, and maintainability.
- Developed the core game logic and player interactions in the model and built an interactive desktop GUI using Java Swing for the View component.
- Managed team tasks and milestones through Git while conducting code reviews and JUnit testing to maintain code quality.